Measured and source-confirmed facts

96 of 96 workers active

Input lock occupied 19.72 s 49.52% of 39.82 s mapping interval

Aggregate mapping concurrency 71.28 equivalent CPUs

Active-worker finish spread 12.93 s

Parsing occurs inside the shared critical section in release source

A02 bounded implementation gate

Corrected full-corpus timing diagnostic complete

Which candidate reduces corrected wall time?

Faster record-safe input preparation

Finer scheduling units SELECTED FOR A02

Bounded producer / consumer queue

No qualifying gain rollback and archive

Measured opportunity, not achieved speed: stages overlap and bottlenecks can move

One logical FASTQ stream per mate

Shared mutexInRead

One worker parses and copies a complete input chunk while locked

Release input lock

That worker maps its private chunk

Another worker may acquire input lock